TS3000 series Robot Controller

TS3000	**SCARA system**
TS3100	**SCARA / 6AXIS system**
TSL3000	**SCARA system**
TSL3100	**6AXIS system**
TS3000E	**SCARA system**
TS3100E	**SCARA / 6AXIS system**
TSL3000E	**SCARA system**
TSL3100E	**6AXIS system**

INSTRUCTION MANUAL

ADDITIONAL FUNCTION MANUAL

Rev.4

September 29, 2020

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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The information contained in this manual is subject to change without notice to effect improvements.

This document explains additional functions of SHIBAURA MACHINE CO.,LTD..'s industrial robot controllers, TS3000/TS3100, TSL3000/TSL3100, and TSL3000E/TSL3100E.

Chapter 1

Explains NCBOY additional axis function.

Chapter 2

Explains the pulse output function.

Chapter 3

Explains the area output function.

Chapter 4

Explains the function for specifying pressing force in newton units.

The information contained in this manual is subject to change without notice.
The respective manuals of Safety, Operation, Robot Language, Interface, Installation and Transportation, Communication, User Parameters, and Alarms are available. Use this manual in combination with the other manuals as appropriate.

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Chapter 1
NCBOY Additional Axis Function

1.1 General description

1.1.1 Features of NCBOY additional axis function

The NCBOY additional axis function combines SHIBAURA MACHINE robot (TH/THL Series, TV/TVL Series) with SHIBAURA MACHINE motor (BS Servo VLBS Series) to expand the robot working range.

Currently, this function can only be used in the vertical articulated robot system. (The SCARA robot system plans to be extended in future.)

The supported versions of the controller systems are as follows:

Main section:	X8GCAV-13F or higher (vertical articulated)
Sequencer section:	X8GCC-06C or higher (TS3000 system) and X8YCC-06C or higher (TSL3000 system)

1.1.2 System configuration

The NCBOY additional axis function is a function that adds SHIBAURA MACHINE motor (BS Servo VLBS Series) SHIBAURA MACHINE robot (TH/THL Series, TV/TVL Series). The usable motor is an absolute encoder type. A resolver type motor cannot be used. The NCBOY additional axis function can add one axis to a TH/THL Series robot and up to two axes to a TV/TVL Series robot. This function provides control of the added motor from a robot operation pendant or a robot program. Adjustment of the added motor (adjustment of the additional axis) is performed using SHANX.

● Addition of one axis

System configuration for adding one axis to a robot is shown in Figure 1.1 and Figure 1.2.

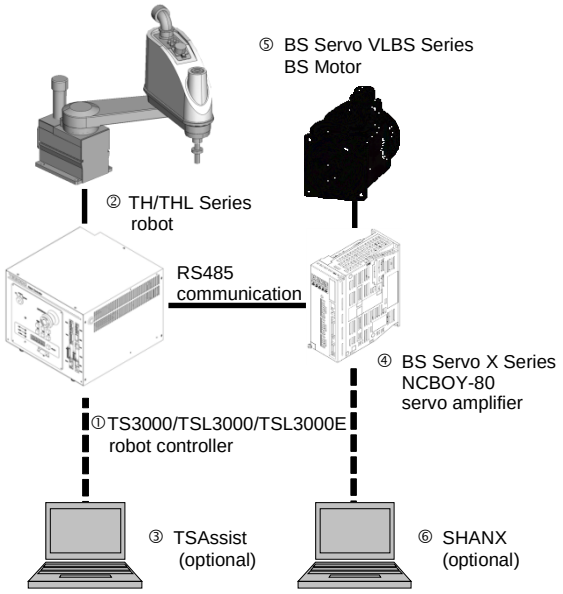


Figure 1.1 Adding one axis to a SCARA robot

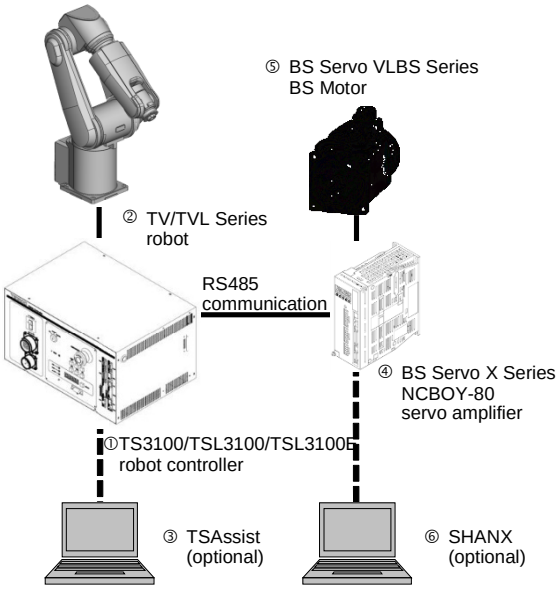


Figure 1.2 Adding one axis to a vertical articulated robot

● Addition of two axes

System configuration for adding two axes to a robot is shown in Figure 1.3.

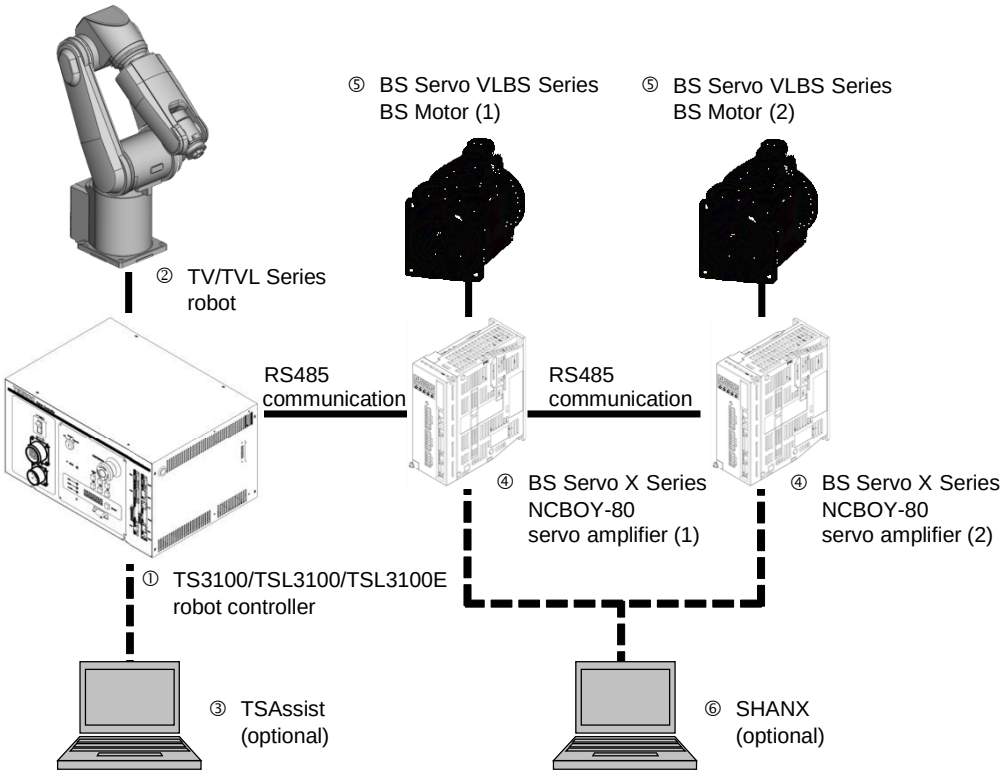


Figure 1.3 Adding two axes to a vertical articulated robot

1.1.3 Installation environment

For installation and installation environment of a robot, robot controller, BS motor, and NCBOY-80, refer to the following documents. In regard to a robot and robot controller, refer to the instruction manual of the model used.

- Robot: Instruction manual, Installation and Transportation
  Installation Environment
  Installation of Robot

- Robot controller: Instruction manual, Installation and Transportation
  Installation of Controller

- BS Motor: Engineering hand book, BS Servo VLBS Series
  1-1. Environment conditions
  1-2. Cautions on installation

- NCBOY-80: Engineering hand book, BS Servo X Series NCBOY-80
  1-1. Cautions on installation
  1-2. Environment conditions

1.2 Specifications

1.2.1 Specifications of NCBOY additional axis function

To a TH/THL Series robot, one axis can be added. To a TV/TVL Series robot, up to two axes can be added.

Specifications of the NCBOY additional axis function are shown in the following table.

Item			Specification
Number of additional axes		SCARA	One
		Vertical articulated	Up to two
Usable controller			TS3000/TS3100/TSL3000/TSL3000E/TSL3100/TSL3100E
Addable servo amplifier			NCBOY-80
Addable motor type			Motor equipped with 17-bit serial absolute encoder (backup using external battery)
Cable between robot and controller		TS3000/TS3100	Up to 25 m
		TSL3000/TSL3100	Up to 10 m
Cable between BS Motor and NCBOY-80			Up to 30 m (restricted by encoder cable specification)
Communication between robot controller and NCBOY-80			RS485, overall length up to 20 m (depending on communication restriction of NCBOY-80)
Ambient conditions	Operating ambient temperature and humidity		Temperature: 0 to 35°C (no freezing) Humidity: 10 to 90%RH (no condensation)
	Operating ambient atmosphere		No dust, metal powder, or corrosive gas
	Storage ambient temperature and humidity		Temperature: -10 to +55°C (no freezing) Humidity: 10 to 90%RH (no condensation)
	Vibration resistance		Conforming to IEC60068-2-6

Addable motor

Rotation speed	Motor type	Output	Motor code UP02	Combined amplifier type
ZA type 3000 min^{-1}	VLBSV-ZA00330S1	30W	02061	VLPSX-008P2
	VLBSV-ZA00530S1	50W	02062	VLPSX-008P2
	VLBSV-ZA01030S1	100W	02063	VLPSX-008P2
	VLBSV-ZA02030S1	200W	02064	VLPSX-008P2
	VLBSV-ZA04030S1	400W	02065	VLPSX-012P2
	VLBSV-ZA06030S1	600W	02066	VLPSX-025P2
	VLBSV-ZA07530S1	750W	02067	VLPSX-025P2
Standard type 1500 min^{-1}	VLBSV-05015S1	500W	02021	VLPSX-012P2
	VLBSV-10015S1	1kW	02022	VLPSX-035P3
	VLBSV-15015S1	1.5kW	02023	VLPSX-035P3
	VLBSV-20015S1	2kW	02024	VLPSX-070P3
	VLBSV-30015S1	3kW	02025	VLPSX-070P3
	VLBSV-50015S1	5kW	02026	VLPSX-100P3
	VLBSV-75015S1	7.5kW	02027	VLPSX-200P3
Standard type 3000 min^{-1}	VLBSV-10030S1	1kW	02041	VLPSX-035P3
	VLBSV-18030S1	1.8kW	02042	VLPSX-070P3
	VLBSV-24030S1	2.4kW	02043	VLPSX-070P3
	VLBSV-30030S1	3kW	02044	VLPSX-070P3
	VLBSV-45030S1	4.5kW	02045	VLPSX-100P3
	VLBSV-70030S1	7kW	02046	VLPSX-200P3
	VLBSV-10K30S1	10kW	02047	VLPSX-200P3

1.2.2 Specifications of system components

For specifications of the robot controller and NCBOY-80, refer to the following materials. In regard to a robot controller, refer to the instruction manual of the model used.

• Robot controller:	• SCARA robot specifications 3. Controller specifications • Vertical articulated robot specifications 3. Controller specifications
• NCBOY-80:	Engineering hand book, BS Servo X Series NCBOY-80 11. Specifications

1.3 Interface

1.3.1 Connection of EXT I/O

EXT I/O wires are used for connection of the NCBOY amplifier in order to increase the number of axes. SHIBAURA MACHINE TS3000/TS3100 controller can be connected with optional TR48DIOCN-1/TR48DIOC-1 or TR48DIOCN-1(L)/TR48DIOC-1(L) module, or SHIBAURA MACHINE remote module. However, only one of these modules can be used when the NCBOY additional axis function is used. When two axes are added to a vertical articulated robot, the above modules cannot be used.

For the communication specification of the EXT I/O wire and NCBOY-80, refer to the following materials. For connection of TR48DIOCN-1/TR48DIOC-1 or TR48DIOCN-1(L)/TR48DIOC-1(L) module, also refer the following materials.

- Robot controller: Instruction manual, Interface EXT I/O
- NCBOY-80: Engineering hand book, BS Servo X Series NCBOY-80 5-6. Mode 24RS485 specifications (TCD protocol slave station)

● Addition of one axis

For the connection, set SW1 of NCBOY-80 as shown in Figure 1.4. Set the station number to 10.

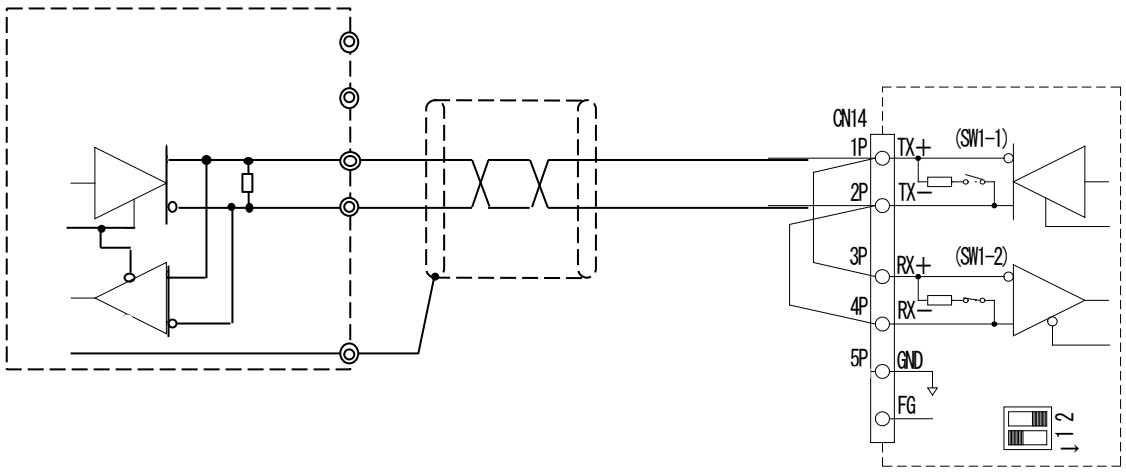


Figure 1.4 Wiring diagram for addition of one axis

● Addition of two axes

For the connection, set SW1 of NCBOY-80 as shown in Figure 1.5. Set the station number to 10 and 12.

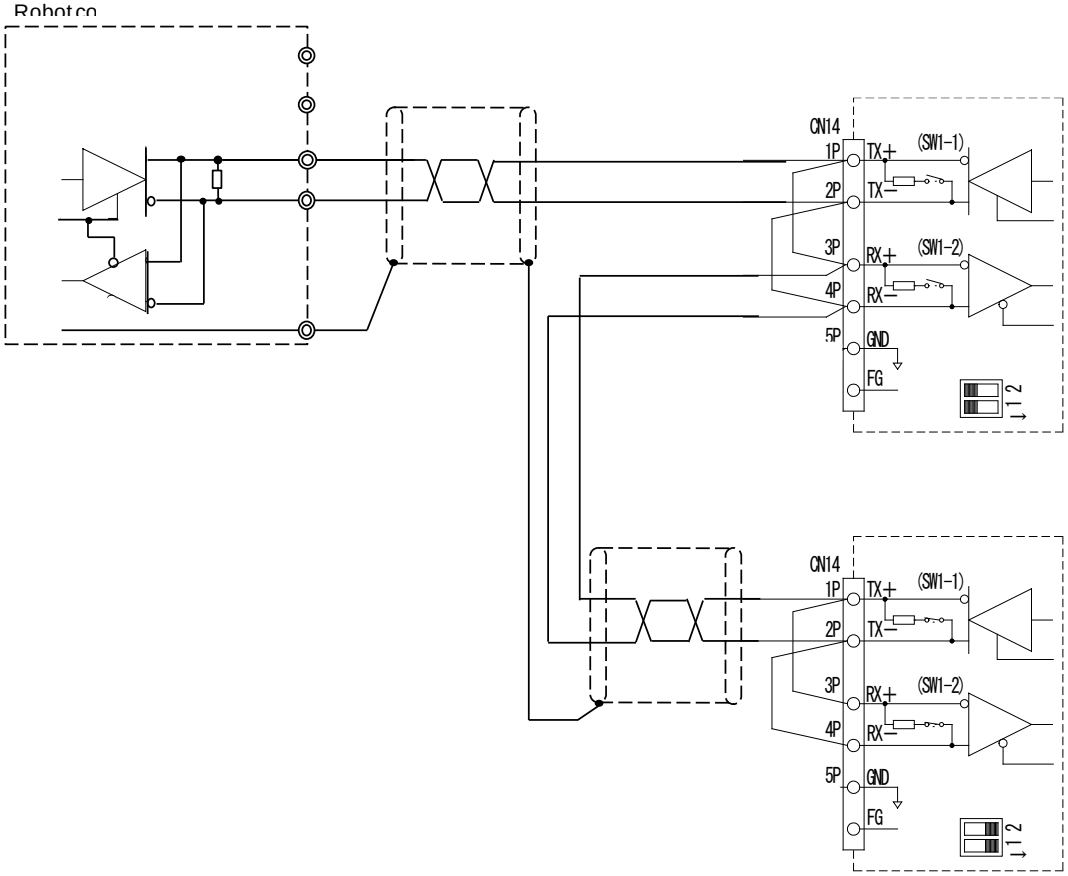


Figure 1.5 Wiring diagram for addition of two axes

● Specifications of connectors and cables

Use twist pair cables for connection. Basically, make series connections. Perform connection according to Figure 1.4 and Figure 1.5. Connect the shield of the twist pair cable to FG of the robot controller.

Before performing wiring or inspection work, wait ten minutes or more after shutting off the power supply, confirm that the charge lamp of NCBOY-80 is turned off, and then check the voltage. Failure to do so may cause electrical shock or a failure of equipment.

The connector terminal block of a robot controller is SL-4000-CWJH-5PGY manufactured by Sato Parts, the connector terminal block of NCBOY-80 is the pluggable terminal block manufactured by Dinkle Enterprise. Use CV14A cables specified by SHIBAURA MACHINE or equivalent products. If a cable with different

specifications is used, not only melting resulted from short circuit or overheating conductor but also a malfunction of the robot resulted from intrusion of noise may be caused. Never use such a cable.

1.3.2 Circuit configuration

For specifications of the power circuit of a robot controller and NCBOY-80, refer to the following materials.

- Robot controller:
  - TS3000 instruction manual, Interface 2 Connection of power supply wire
  - TSL3000 instruction manual, Interface 2 Power supply wire
- NCBOY-80: Engineering hand book, BS Servo X Series NCBOY-80 2-1. Connection of power circuit

For connection of a robot controller with NCBOY-80, refer to Figure 1.6.
Figure 1.6 shows an example of connection with VLPSX-008P. Some NCBOY-80 have the retaining brake or the dynamic brake. Before perform wiring, confirm the amplifier used.

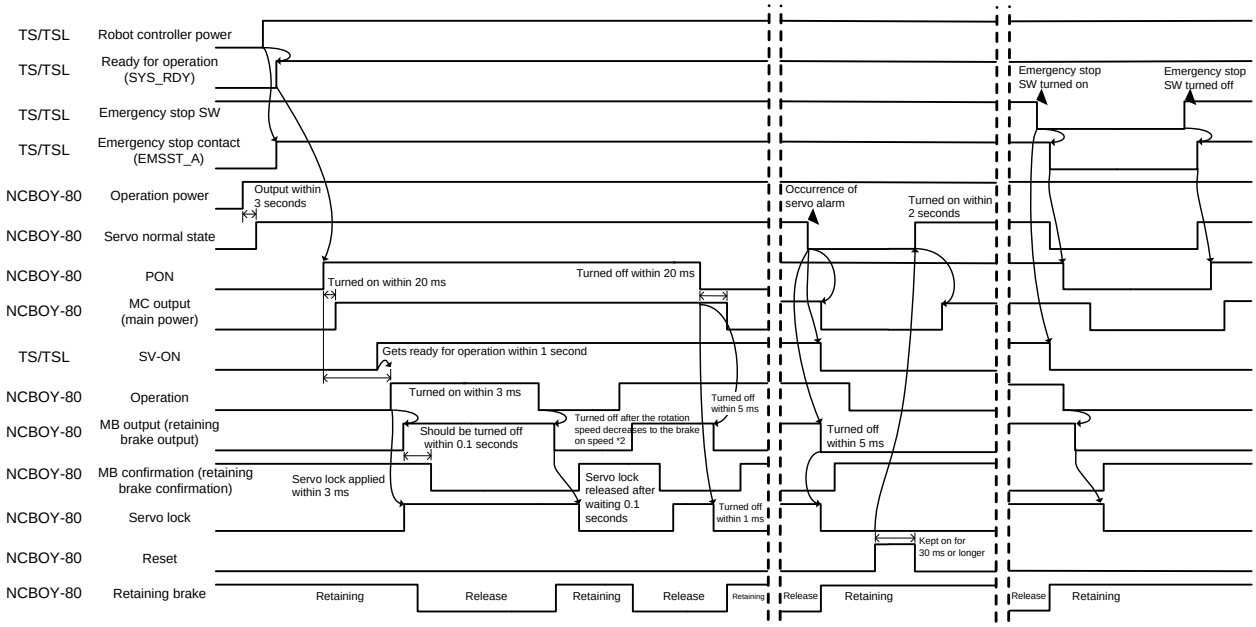


Figure 1.6 Sequence diagram

● TS3000/TS3100 wiring diagram

Figure 1.7 shows the wiring diagram between TS3000/TS3100 (Type N) robot controller and NCBOY-80 (VLASX-008P).

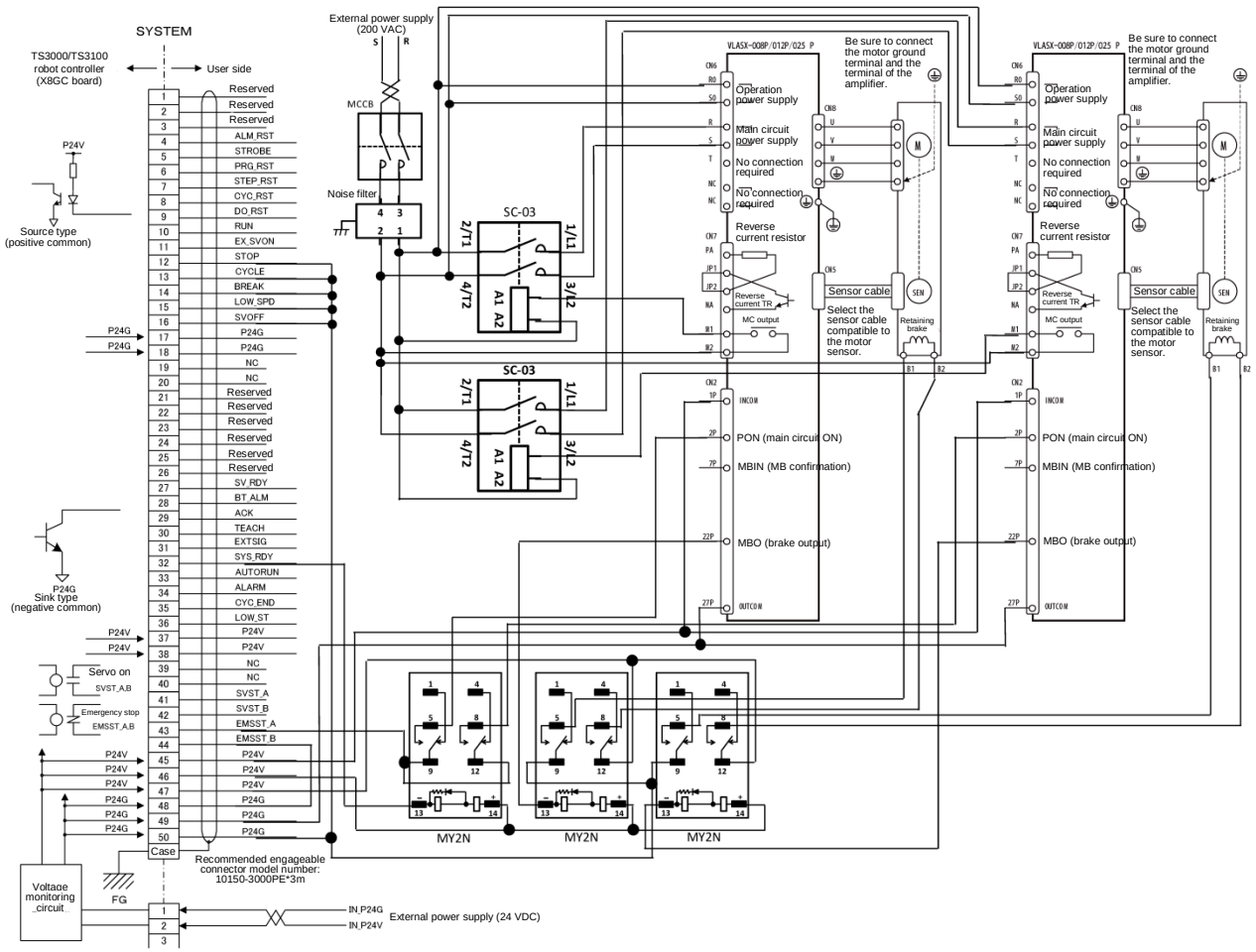


Figure 1.7 Wiring diagram (between TS3000/TS3100 (Type N) and VLASX-008P/012P/025P)

*Note 1 Wire the power supply and motor armature wires separately from the motor sensor cable.

*Note 2 Connect the motor ground wire to the ground terminal of the servo amplifier to ensure grounding of the servo amplifier.

*Note 3 Provide the contactor and the relay coils with surge killers as a measure against noise.

*Note 4 This description is based on an assumption that the TS3000/TS3100 controller is a type N.

System configuration table (TS3000/TS3100)

Name	Type	Qty.	Manufacturer
Robot controller	TS3000/TS3100	1	SHIBAURA MACHINE
NCBOY-80	VLPSX-008P	2	SHIBAURA MACHINE
BS Motor	VLBSV-ZA02030S1	2	SHIBAURA MACHINE
Breaker MCCB	SA32C-3A	1	Fuji Electric
Safety relay	MY2N-2D DC24	3	OMRON
Thermal relay	SC-03 1a AC200V	2	Mitsubishi Electric
Noise filter	PTHN-5006	1	TDK
I/O signal cable	CV02C-101A	2	Recommended product
ZA serial ABS cable	CV05E-101A	2	Recommended product
Single phase power supply cable	CV06A-101B	2	Recommended product
Built-in reverse current absorption resistor and MC cable	CV07A-101B	2	Recommended product
Armature cable for ZA motor equipped with brake	CV08B-101A	2	Recommended product
Connector terminal 36-pin	PCN-10S36	2	Sumitomo 3M
Connector terminal 50-pin	PCN-10S50	1	Sumitomo 3M

* The above system configuration table shows peripheral equipment necessary for connection of TS3000/TS3100 (N type) robot controller with NCBOY-80 (VLASX-008P).
  The amplifier and system configuration depends on the capacity of the BS motor used. Therefore select peripheral equipment according to the following material.

• NCBOY-80: Engineering hand book, BS Servo X Series NCBOY-80
  2-2. Selection of peripheral equipment

● TSL3000/TSL3100 wiring diagram

Figure 1.8 shows the wiring diagram between TSL3000/TSL3100 (Type N) robot controller and NCBOY-80 (VLASX-008P).

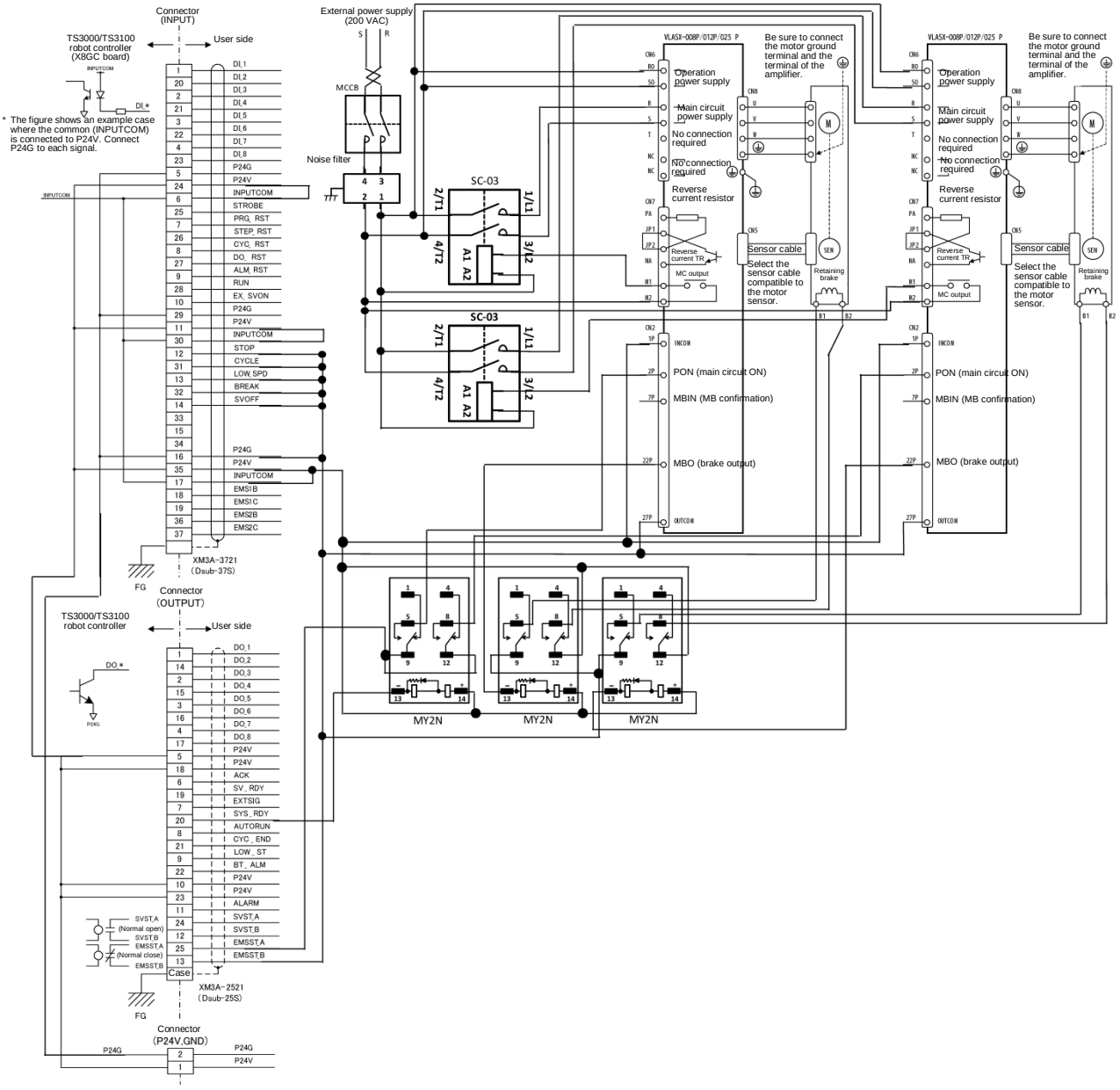


Figure 1.8 Wiring diagram (between TSL3000/TSL3100 (Type N) and VLASX-008P/012P/025P)

*Note 1 Wire the power supply and motor armature wires separately from the motor sensor cable.

*Note 2 Connect the motor ground wire to the ground terminal of the servo amplifier to ensure grounding of the servo amplifier.

*Note 3 Provide the contactor and the relay coils with surge killers as a measure against noise.

*Note 4 This description is based on an assumption that the TS3000/TS3100 controller is a type N.

System configuration table (TSL3000/TSL3100)

Name	Type	Qty.	Manufacturer
Robot controller	TSL3000/TSL3100	1	SHIBAURA MACHINE
NCBOY-80	VLPSX-008P	2	SHIBAURA MACHINE
BS Motor	VLBSV-ZA02030S1	2	SHIBAURA MACHINE
Breaker MCCB	SA32C-3A	1	Fuji Electric
Safety relay	MY2N-2D DC24	3	OMRON
Thermal relay	SC-03 1a AC200V	2	Mitsubishi Electric
Noise filter	PTHN-5006	1	TDK
I/O signal cable	CV02C-101A	2	Recommended product
ZA serial ABS cable	CV05E-101A	2	Recommended product
Single phase power supply cable	CV06A-101B	2	Recommended product
Built-in reverse flow absorption resistor and MC cable	CV07A-101B	2	Recommended product
Armature cable for ZA motor equipped with brake	CV08B-101A	2	Recommended product
Connector terminal 36-pin	PCN-10S36	2	Sumitomo 3M
Connector terminal 37-pin	PCN-10S37	1	Sumitomo 3M
Connector terminal 40-pin	PCN-10S40	1	Sumitomo 3M

* The above system configuration table shows peripheral equipment necessary for connection of TSL3000/TSL3100 (N type) robot controller with NCBOY-80 (VLASX-008P).
  The amplifier and system configuration depends on the capacity of the BS motor used. Therefore select peripheral equipment according to the following material.

• NCBOY-80: Engineering hand book, BS Servo X Series NCBOY-80
2-2. Selection of peripheral equipment

1.4 NCBOY parameter setting

1.4.1 TS3000 parameter

[U48] NCBOY ADDITIONAL AXIS

```
[U48] NCBOY ADDITIONAL AXIS
{Use/Not Use} (0:Not Use, 1:Use )
= 0  0
```

Setting: = (Enabling or disabling NCBOY additional axis 1) (Enabling or disabling NCBOY additional axis 2)
Data type: Integer type
Unit of data: None
Data range: 0 or 1
Number of data: 2
Description: This parameter enables or disables the NCBOY additional axis function.
* Only one additional axis can be added to the SCARA robot.

[U49] NCBOY AXIS INCHING AMOUNT

```
[U49] NCBOY AXIS INCHING AMOUNT
{ Joint Coordinate  [mm][deg]}
= 1.0     1.0
= 0.5     0.5
= 0.1     0.1
```

Setting: = (NCBOY additional axis 1 large inching amount) (NCBOY additional axis 2 large inching amount)
= (NCBOY additional axis 1 medium inching amount) (NCBOY additional axis 2 medium inching amount)
= (NCBOY additional axis 1 small inching amount) (NCBOY additional axis 2 small inching amount)
Data type: Real number type
Unit of data: mm or deg
Data range: 0.0001 to limit value
Number of data: 2 * 3 (large, medium, small)
Description: This parameter specifies the inching amount of the NCBOY additional axis for operation of joint guide from the teaching pendant of robot controller.
* Only one additional axis can be added to the SCARA robot.

1.4.2 NCBOY parameter

The parameter settings for connection of the controller and the BS amplifier of the NCBOY connecting function are shown in the following table.
Parameters not described in the following table (such as motor adjustment) should be set as necessary.

Symbol	Name	Set value	Description and note
UP01	Control mode	24	RS485 (TCD)
UP02	Motor code		Refer to 1.2.1 NCBOY additional axis function specifications.
UP04	Electronic gear numerator		Set the electronic gear so that one pulse corresponds to one micrometer at the current position.
UP05	Electronic gear denominator		
An	Station number	10 or 12	Set this to 10 for the additional axis 1 and 12 for the additional axis 2.
PP11	Feed rate 0	100	Specify the speed in a program operation.
PP12	Feed rate 1	10	Specify the speed in an inching operation.
PP13	Feed rate 2		Not used
PP14	Feed rate 3		Not used
PP15	JOG speed 0		Not used
PP16	JOG speed 1	10	Set the speed HIGH of TS3000 JOG operation.
PP17	JOG speed 2	5	Set the speed MID of TS3000 JOG operation.
PP18	JOG speed 3	1	Set the speed LOW of TS3000 JOG operation.
PP31	Sequence input invert	100	Set the over travel input and the dynamic brake confirmation input to "Disable". To use the over travel input or the dynamic brake confirmation input, set this parameter according to the input logic.
PP37	Acceleration time 1	100	Set the acceleration and deceleration time (msec) for the program operation, JOG operation, and inching operation. Set the time in accordance with the specifications of the additional axis used.
PP38	Deceleration time 1	100	
PP39	Acceleration time 2		Not used
PP40	Deceleration time 2		
PP41	Acceleration time 3		
PP42	Deceleration time 3		
PP43	Acceleration time 4		
PP44	Deceleration time 4		
PP45	Baud rate setting	7	Set the RS485 communication speed to 250 Kbps.
PP48	RS485 setting	0000	Set this parameter to "No parity" and "Stop bit 1".
PP59	Start confirmation signal function	1	This parameter is used for catching the start signal without fail in cases such as positioning with small travel amount. Failure to set this parameter may cause positioning malfunction, so be sure to set this parameter.

1.5 Control and operation

1.5.1 Current position display

On a SCARA robot system, the current position of the NCBOY additional axis is displayed as the axis 5 of the JOINT/WORLD position.
On a vertical articulated robot system, the current position of the NCBOY additional axis is displayed as the axis 7 and the axis 8 of the JOINT/WORLD position.

1.5.2 Servo ON, Servo OFF, Alarm reset, and Emergency stop

When operation of servo ON, servo OFF, or alarm reset is performed on TS3000, the same operation is performed also on NCBOY-80.
For emergency stop operation, the emergency stop contact output of the robot controller is connected to NCBOY-80 using an exclusive wire.
Because of difference of design philosophy between the robot controller and NCBOY-80, the PON (main circuit) is turned ON and the power (200 V) is supplied to the main circuit when the emergency stop button is released on the robot. When an emergency stop is turned ON (an emergency stop state is activated), the PON is turned OFF and the main circuit is interrupted.

1.5.3 JOG operation

- SCARA — JOG operation for using [5+] and [5-] of the axis 5 of teaching pendant to operate the additional axis.
- Vertical articulated — JOG operation for using [1+] and [1-] of the axis 7 and [2+] and [2-] of the axis 8 of teaching pendant to operate the additional axis.

Set the JOG speed in PP16, PP17, and PP18 of NCBOY corresponding to the specified teaching pendant speed HIGH, MID, and LOW.
For inching, use an INC feeding amount command of NCBOY to perform feeding of specified increment amount.
(The specified feeding amount is specified by the parameter of the robot controller.)
For the inching speed, set the speed in PP12 of NCBOY-80.

1.6 Programming

Additional language

MOVEN 7, Target coordinate value

INPOS7	State of NCBOY additional axis	In-position (1: stopping/0: Running)
INPOS8	State of NCBOY additional axis	In-position (1: stopping/0: Running)
INPOS5	State of NCBOY additional axis	In-position (1: stopping/0: Running)

Program example

```
GLOBAL
    POINT1 =100.0
    POINT2 =200.0
END
PROGRAM MAIN
    MOVE P1                          Position the robot.
    MOVEN 7,100.000                  Move the axis 7 to 100.000.
    WAIT INPOS7==1                   Wait for finish of operation of the axis 7.
    MOVE P2                          Position the robot.
    MOVEN 7,POINT1                   Move the axis 7 to 100.0.
    MOVE P3                          Position the robot (not waiting for finish
                                     of operation of the axis 7)
    WAIT INPOS7==1                   Wait for finish of operation of the axis 7.
    MOVE P4                          Position the robot
    MOVEN 8,POINT2 WITH SPEED=50     Move the axis 8 to 200.0 at 50% of the
                                     maximum speed.
    WAIT INPOS8==1                   Wait for finish of operation of the axis 8.
END
```

SPEED command

A SPEED command is used in a same manner as the conventional language, and can be used in a WITH clause.

OVERRIDE command

Also an OVERRIDE command is used in a same manner as the conventional language.

The OVERRIDE is always changing. Therefore it is updated by interruption every 3 msec.

1.7 Error information

① Alarms that occur when the NCBOY additional axis function is enabled and communication has not been established:

8-101 NCBOY Axis5 Communication (SCARA, additional axis 1)
8-103 NCBOY Axis7 Communication (6-axis, additional axis 1)
8-104 NCBOY Axis8 Communication (6-axis, additional axis 2)

② When communication is established and an alarm is notified from NCBOY-80 using three bits of X000, X00A, and X00B. An alarm occurs as the followings depending on the bit informing the alarm.

X000: Servo abnormality	8-097	NCBOY AXIS5 ALARM (SCARA, additional axis 1)
	8-099	NCBOY AXIS7 ALARM (6-axis, additional axis 1)
	8-100	NCBOY AXIS8 ALARM (6-axis, additional axis 2)
X00A: Low battery	1-069	NCBOY AXIS5 BATTERY LOW (SCARA, additional axis 1)
	1-071	NCBOY AXIS7 BATTERY LOW (6-axis, additional axis 1)
	1-072	NCBOY AXIS8 BATTERY LOW (6-axis, additional axis 2)
X00B: Warning	1-065	NCBOY AXIS5 WARNING (SCARA, additional axis 1)
	1-067	NCBOY AXIS7 WARNING (6-axis, additional axis 1)
	1-068	NCBOY AXIS8 WARNING (6-axis, additional axis 2)

③ When alarm resetting is performed on the robot controller, the alarm on NCBOY-80 is cleared.

1.8 Caution

- Be aware that when operation of an additional axis is interrupted by an emergency stop, the operation of the additional axis will not be resumed by a later RUN command.
- Be aware that when a SEGMENT operation is performed, a MOVEN command cannot stop the operation.
- Do not implement a MOVEN command during ENABLE PASS.

Chapter 2
Pulse Output Function

2.1 General description

The pulse output function is a function that outputs the robot fingertip speed as a pulse signal to external equipment.

The pulse output is used for monitoring the fingertip speed of a robot.

The controller system versions in which this function is supported are as follows:

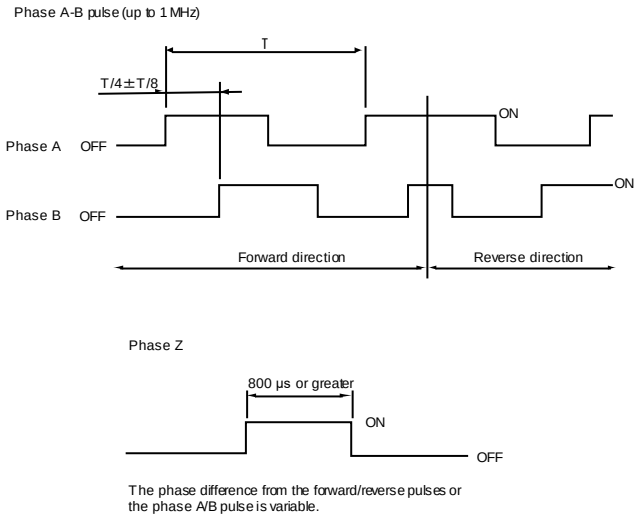
Main section: X8GCAS-11F or higher (SCARA) and X8GCAV-11K or higher (vertical articulated)

Servo section: X8GLA-11A or higher (TS3000 system) and X8YCB-11A or higher (TSL3000 system)

Note that the pulse output can be used when the version of the main board of the controller is X8GLA-2 or higher (board with a pulse output circuit added).

2.2 Pulse output specifications

This section explains specifications of the pulse output function.

Item	Specifications	Note
1. Target model	TS3000/TS3100	Available as an optional function
2. Pulse accuracy	Command: 0.07% • OVERRIDE: 100% • Pulse Weight: 1.0 Feedback: 3.5% • OVERRIDE: 100% • Pulse Weight: 1.0	
3. Pulse and trigger signal interface	Supply: ±3.3 V (supplied internally) Method: A, B, Z 90 deg. phase difference Specification: RS422 differential output	
4. Differential output specifications	Maximum output frequency: 4 Mpps AB phase pulse: 1 MHz  The phase difference from the forward/reverse pulses or the phase A/B pulse is variable.	

2.3 Fabrication of cable

To fabricate a cable, follow the instructions below. Use the supplied connector [PCR-E20FS+ (connector), PCR-V20LB (hood)].

① Circuit configuration

Use a shield twist pair cable as shown in the following figure. Make a connection in pairs as the following figure. Note that incorrect connection causes deterioration of the noise resistance performance and may lead to a malfunction.

* Connect the cable shield to the pin 10 of the half pitch connector.

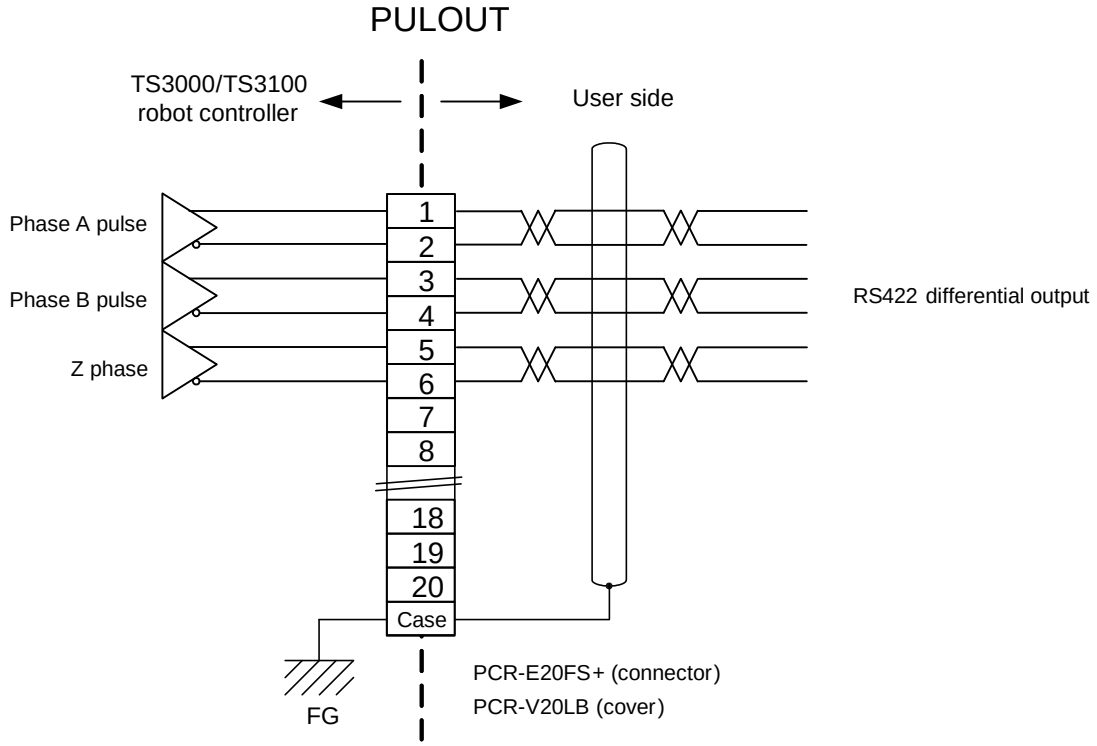


Figure 2.1 Pulse output function circuit connection diagram

② Specifications of connector and cable

• Number of cable core wires:	AWG 28 (max.)
• Standard conformable external diameter:	ϕ8
• Shield:	Connected to pin 10 of half pitch connector
• Cable length:	up to 10 m

If a cable with different specifications is used, not only melting resulted from short circuit or overheating conductor but also a malfunction of the robot resulted from intrusion of noise may be caused. Never use such a cable.

③ Cable fabrication method

The connector is a solder cup type to which wires is connected by soldering.

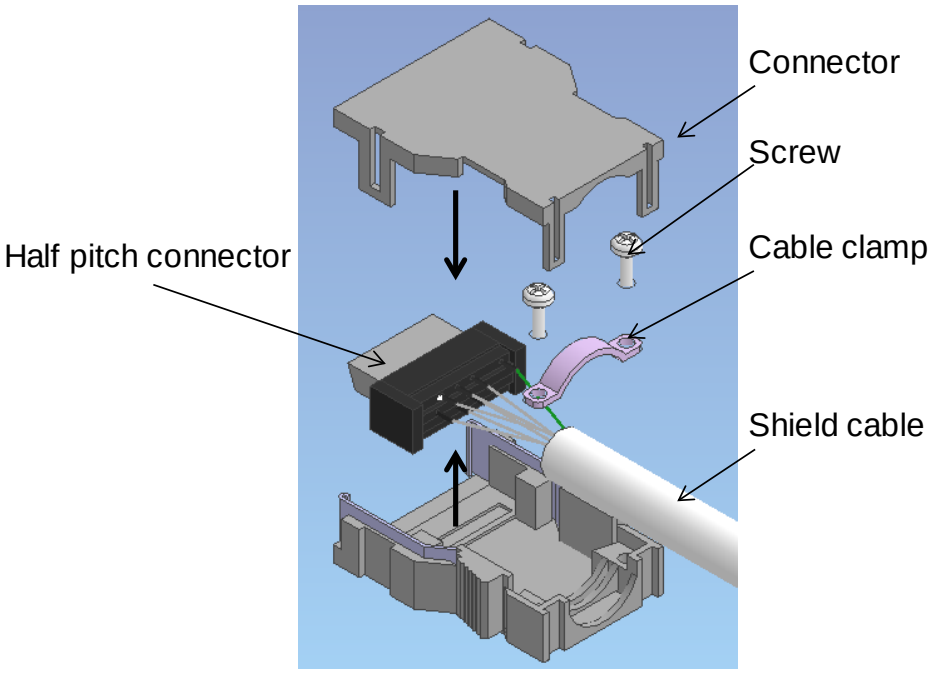


Figure 2.2 Cable fabrication method

(Note) The shield of a cable is connected through the housing to the grounding of TS3000 robot controller.

2.4 Parameter

The setting parameter for the pulse output function is described in "[U46] PULSE OUTPUT FUNCTION" of the "USER.PAR" user parameter file. The parameter format is as the followings.

```
[U46] PULSE OUTPUT FUNCTION
{0:Disabled 1:Feedback mode 2:Command mode}
= 0                    ①Pulse data type
{ Pulse Weight}
= 1.0                  ②Pulse weight per travel amount of robot
                         fingertip
{ I/O Number For Trigger [1-64]}
= 1                    ③Trigger output
```

① Pulse data type

This parameter specifies the type of output pulse data.
When {Feedback mode} is set, the pulse to be output is based on the robot fingertip speed calculated from the motor feedback value. When {Command mode} is set, the pulse to be output is based on the speed commanded to the robot. When the pulse output function is to be unused, set this parameter to {Disabled}.

* The default setting of this parameter is {Disabled}.

Data type: Integer type
Data range: 0 to 2
Value example: 0

② Pulse weight per travel amount of robot fingertip

This parameter specifies the pulse weight [pulse] per travel amount of robot fingertip [μm].

* The frequency of the pulse output function is 3 ms. Therefore the value set by this parameter is actually rounded off to a multiple of the calculation frequency, 3 ms.

<Note>
The default setting of this parameter for the pulse output function is 1.0 [pulse]. This parameter can be set to a larger value, but note the following explanations for adjustment of the value.

■ When this parameter is set to a smaller value:
A calculation error of the robot fingertip speed can be absorbed.
■ When this parameter is set to a larger value:
The response of the pulse amount to a change in the robot fingertip speed becomes higher.

Data type: Real number type
Unit of data: pulse/μm
Data range: 0.0 and higher
Value example: 1.0

③ Trigger output
One trigger output function is originally equipped, separately from “TRIG” described in Instruction Manual: Interface Manual.
The pins 5 and 6 of the "PULOUT" half pitch 20-pin connector that is located on the upper part of the TS3000/TS3100 robot controller backside are used for the trigger output (RS-422 differential output). To activate the trigger output, use a "DOUT" command of the SCOL language.

* This output is common to the "DOUT" of the conventional external output signal wire. When using this output, pay attention not to allow overlapping. If the device is used with the output overlapped, a malfunction may be caused.

Data type: Integer type
Data range: 1 to 64
Value example: 1

2.5 Programming

This section shows a sample program of the pulse output function.

```
PROGRAM PULSESMPLE
    ENABLE NOWAIT                   'Enables the system switch.
    MOVE START
    MOVE A + POINT(0,0,50,0,0,0)
    ACCUR = FINE                    'Sets the robot positioning accuracy to
                                    "Fine".
    WAIT MOTION >= 100              'Waits for finish of the latest operation.
    DOUT(1)                         'Outputs a trigger of the pulse output
                                    function.
    MOVE A
    MOVEC B C
    ACCUR = COARSE                  'Sets the robot positioning accuracy to
                                    "Coarse".
    MOVE C + POINT(0,0,50,0,0,0)
    MOVE FIN
    DISABLE NOWAIT                  'Disables the system switch.
END
```

Chapter 3
Area Output Function

This chapter explains the area output function.

3.1 Overview of the area output function

The area output function is a function for turning on or off a specified DOUT signal when the robot enters a specified area. It allows you to choose 64 combinations of a specified area and a specified DOUT signal.
The controller system versions in which this function is supported are as follows:
Main section: X8GCAS-13E or higher (SCARA) and X8GCAV-13C or higher (vertical articulated)

3.2 Parameters

The setting parameters for the area output function are described in the “AREA.PAR” parameter file. The parameter format is as follows:

```
[A00] AREA OUTPUT SETTING PARAMETER
{Area output flag (0:Disable 1:Enable)}
{P1:+Direction position coordinate [mm]}
{P2:-Direction position coordinate [mm]}
{Dout number}
{Table No.1}
=   0           ①Area output function enabling/disabling setting
=   9999.9   9999.9   9999.9   0.0   0.0   0.0   0.0   0.0
=  -9999.9  -9999.9  -9999.9   0.0   0.0   0.0   0.0   0.0
               ②Cuboid areas P1 and P2
=   1           ③Output DOUT number and output logic
{Table No.2}
=   0
=   9999.9   9999.9   9999.9   0.0   0.0   0.0   0.0   0.0
=  -9999.9  -9999.9  -9999.9   0.0   0.0   0.0   0.0   0.0
=   2
{Table No.3}
=   0
=   9999.9   9999.9   9999.9   0.0   0.0   0.0   0.0   0.0
=  -9999.9  -9999.9  -9999.9   0.0   0.0   0.0   0.0   0.0
=   3

.
.
.

{Table No.64}
=   0
=   9999.9   9999.9   9999.9   0.0   0.0   0.0   0.0   0.0
=  -9999.9  -9999.9  -9999.9   0.0   0.0   0.0   0.0   0.0
=   64
```

① **Area output function enabling/disabling setting**

This parameter specifies whether to enable or disable the area output function.

0: Disables the function

1: Enables the function

Data type: Integer type

Data range: 0 or 1

Value example: 0

② **Cuboid areas P1 and P2**

These parameters specify an area for which a signal is output. When the robot enters the specified cuboid area, the specified DOUT signal is turned on or off.

* If the values of (plus position) and (minus position) are interchanged, this does not affect operation.

Setting: = (plus position x-coordinate), (plus position y-coordinate), (plus position z coordinate), (reserved), (reserved), (reserved), (reserved), (reserved)

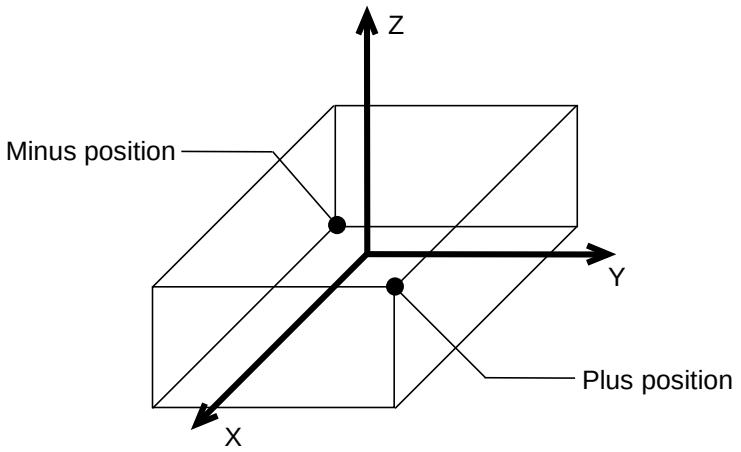
=(minus position x-coordinate), (minus position y-coordinate), (minus position z coordinate), (reserved), (reserved), (reserved), (reserved), (reserved)

Data type: Real number type

Data unit: mm

Data range: -9999.9 to 9999.9

Value example: 1.0



③ **Output DOUT number and output logic**

This parameter specifies the DOUT number for which a signal is output.
If the DOUT number is positive, the specified DOUT is turned on when the robot enters the specified area. Also, if the DOUT number is negative, the specified DOUT is turned off when the robot enters the specified area.

Data type: Integer type
Data range: 1 to 64, -1 to -64, 101 to 164, -101 to -164, 301 to 364, -301 to -364, 401 to 464, -401 to -464
Value example: 1

Note:

- If the same DOUT number is specified on more than one table, the output signal will be the logical OR of the DOUT output of each table. Details are provided in the next section.
- Note that a specified DOUT enabled by this function cannot be turned on or off by using the SCOL language or a teaching pendant.

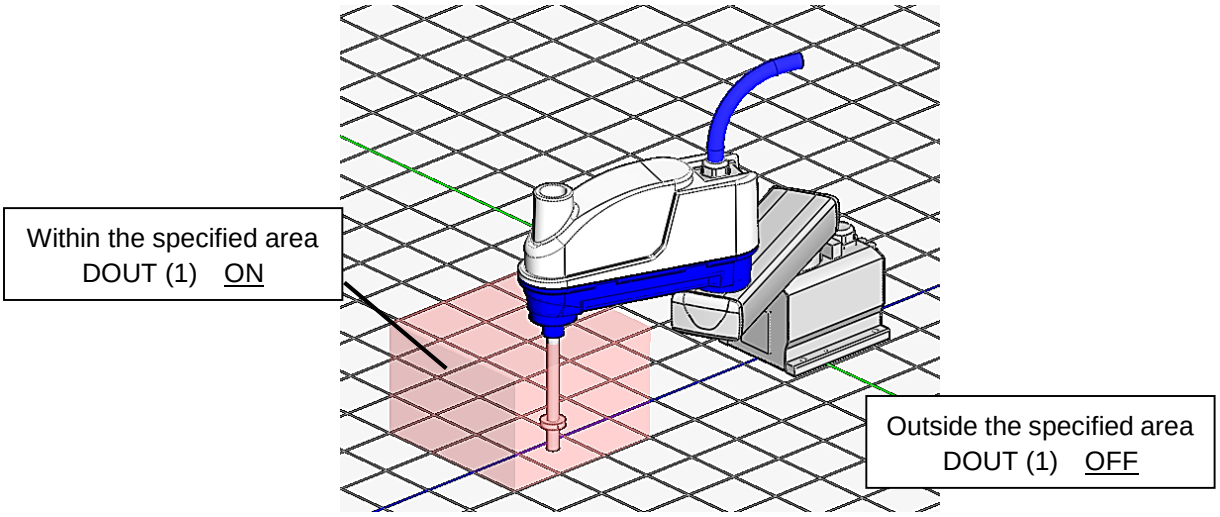
3.3 Usage Example

This section gives an example where AREA.PAR is written as follows:

```
[A00] AREA OUTPUT SETTING PARAMETER
{Area output flag (0:Disable 1:Enable)}
{P1:+Direction position coordinate [mm]}
{P2:-Direction position coordinate [mm]}
{Dout number}
{Table No.1}
=    1
=    700.0    100.0    200.0    0.0    0.0    0.0    0.0    0.0
=    400.0   -200.0      0.0    0.0    0.0    0.0    0.0    0.0
=    1

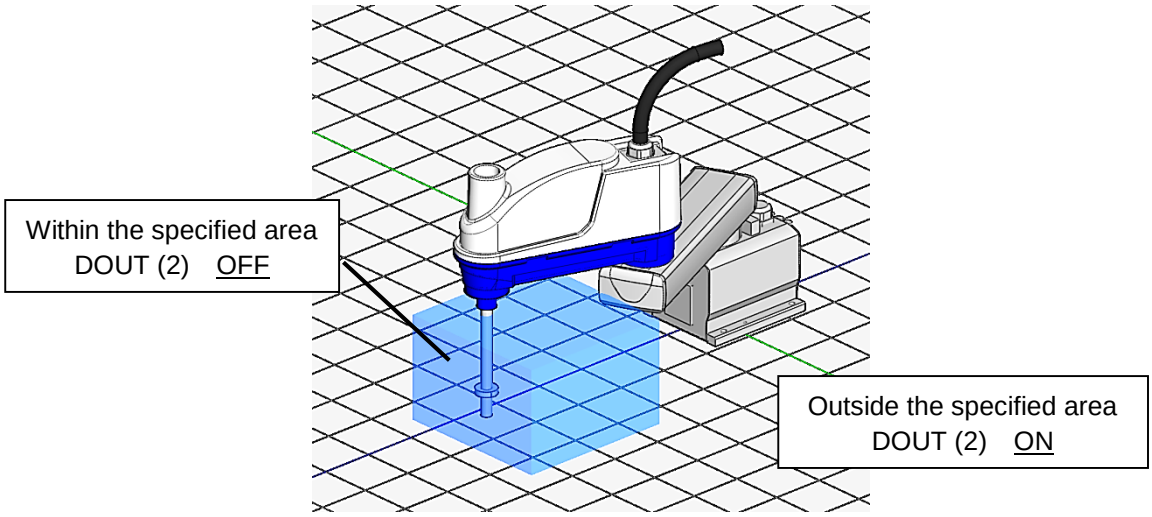
{Table No.2}
=    1
=    650.0    250.0    200.0    0.0    0.0    0.0    0.0    0.0
=    350.0    -50.0      0.0    0.0    0.0    0.0    0.0    0.0
=    -2
.
.
.
```

The area specified in Table No.1 is as shown in the figure below.



If the current position of the robot is within the specified area, DOUT (1) is turned on; otherwise, DOUT (1) is turned off.

The area specified in Table No.2 is as shown in the figure below.



If the current position of the robot is within the specified area, DOUT (2) is turned off; otherwise, DOUT (2) is turned on.

3.3.1 Case where the same DOUT number is used on more than one table

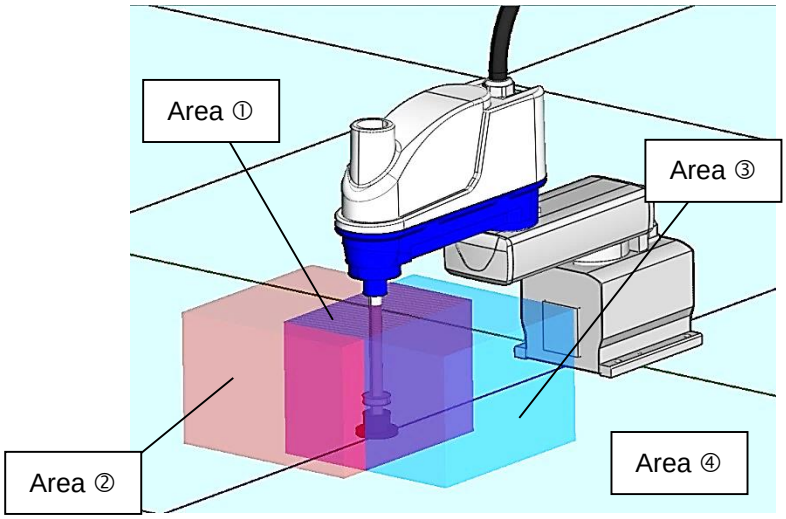
If this area output function is used to specify the same DOUT number on more than one table, the output signal will be the logical OR of the DOUT output of each table. In other words, if output ON and OFF instructions are given at the same time, the ON instruction will be given priority.

Here is an example where AREA.PAR is written as follows:

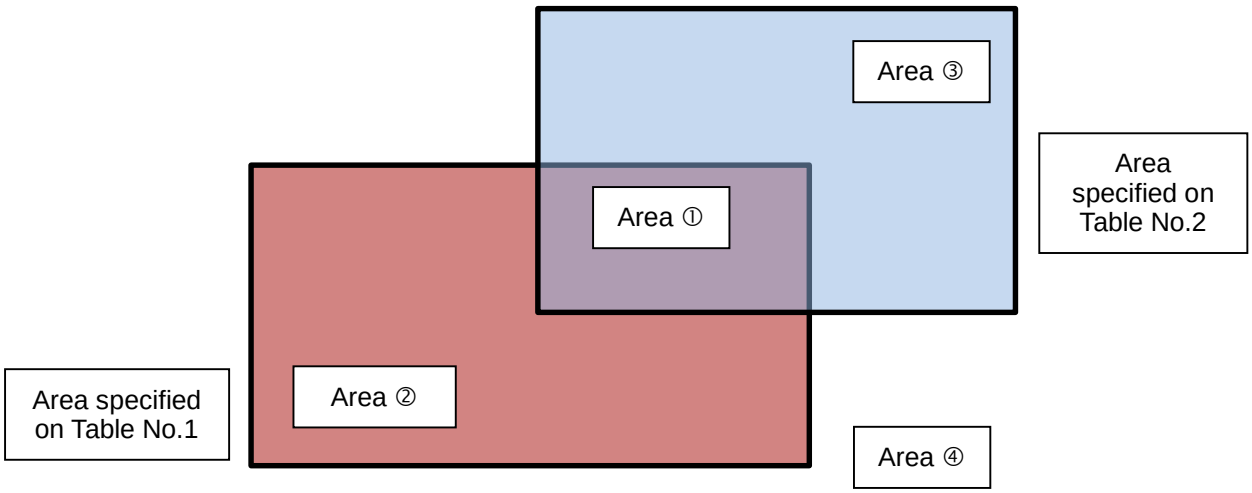
```
[A00] AREA OUTPUT SETTING PARAMETER
{Area output flag (0:Disable 1:Enable)}
{P1:+Direction position coordinate [mm]}
{P2:-Direction position coordinate [mm]}
{Dout number}
{Table No.1}
=    1
=    700.0    100.0    200.0    0.0    0.0    0.0    0.0    0.0
=    400.0   -200.0      0.0    0.0    0.0    0.0    0.0    0.0
=    1

{Table No.2}
=    1
=    650.0    250.0    200.0    0.0    0.0    0.0    0.0    0.0
=    350.0    -50.0      0.0    0.0    0.0    0.0    0.0    0.0
=    -1
.
.
.
```

The specified areas are as shown below.



Also, the areas are divided into the following four areas depending on the combination of the setting values of each table.



ON/OFF of DOUT (1) in each area is as follows:

Area ①: Table No.1 is ON, Table No.2 is OFF, and DOUT (1) is ON.
Area ②: Table No.1 is ON, Table No.2 is ON, and DOUT (1) is ON.
Area ③: Table No.1 is OFF, Table No.2 is OFF, and DOUT (1) is OFF.
Area ④: Table No.1 is OFF, Table No.2 is ON, and DOUT (1) is ON.

Chapter 4
Function for Specifying Pressing Force in Newton Units

This chapter describes the function for specifying the pressing force in newton units.

4.1 Overview of function for specifying the pressing force in newton units

The function for specifying the pressing force in newton units enables the pressing force applied by the SCARA robot in the Z-axis direction to be specified in newton (N) units instead of the regular percentage (%) units.

Models: THE400 / TSL3000,
THL600 / TSL3000, THL600 / TS3000
Main unit: X8GCAS-14B or later

* This function is enabled only when [R45] FORCE3LIM FUNCTION in ROBOT.PAR is enabled.

4.2 Language specifications

The command FORCE3LIM is used in the function for specifying the pressing force in newton units. The language details are as follows.

FORCE3LIM

Function	Specifies the torque limit value of the 3rd axis in Nm units.
Format	FORCE3LIM = (Statement)
Examples	FORCE3LIM=50 MOVE A1 WITH FORCE3LIM=70
Exaplanation and notest	Specifies the torque limit value of the 3rd axis for the SCARA robot in Nm units. Constants, variables, and calculation formulas can be used in the Statement section. If a value less than or equal to 0 is specified as the torque limit value, it is treated as 0. If the torque limit value is too low, the robot may be unable to operate, and an operation error may occur. If a value larger than the torque value that can be output is specified, the maximum torque that can be output is set. For example, when the output torque of the 3rd axis is 50%=75 Nm, TORQUE={300,300,50,300,300} and FORCE3LIM=75 will yield the same results. Internally, the 3rd axis setting locations for the FORCE3LIM and TORQUE commands are identical. Therefore, after a torque limit is applied by FORCE3LIM=75, the limit value can be changed using the TORQUE command to TORQUE={300,300,300,300,300}.

Sample Program

```
PROGRAM TORQUESMPL
MOVE A1              Z-axis (3rd axis in this case) torque
FORCE3LIM=75         during motion to A2 is limited to a rated
MOVE A2              torque of 75 Nm.
OPEN1
DELAY 0.5
MOVE A1
FORCE3LIM=999        Torque limit is set to a maximum value.
READY                (Torque limit disabled)
END
```

4.3 Changing the torque display units in teaching pendant

The 3rd axis torque display units for "UTILITY" →" MOTOR" in the teaching pendant can be changed from % to N.
[U51] in USER.PAR is used to set the 3rd axis torque display units to the conventional percentage (%) or to newtons (N).

```
[U51] 3-AXIS TORQUE DISPLAY (SCARA)
{0:[%] 1:[N] (Add Payload) 2:[N] (Not Add Payload)}
= 0
```

The default is 0.
0: The torque is displayed as the conventional percentage (%).
1: The torque is displayed in newtons (N). (This includes the mass from PAYLOAD.)
2: The torque is displayed in newtons (N). (This does not include the mass from PAYLOAD.)

* The torque display units change function is enabled when [R45] FORCE3LIM FUNCTION in ROBOT.PAR is enabled. If [R45] is disabled, the torque is displayed as a percentage (%) even if 1 or 2 is set in [U51].

* The TP3000 teaching pendant also supports this torque display units change function.
  However, this support is available only for TP3000 system software versions X8KBA-02E and later.